HVDC Design

AmWatt Team Presenation

Last Update: September 30, 2026

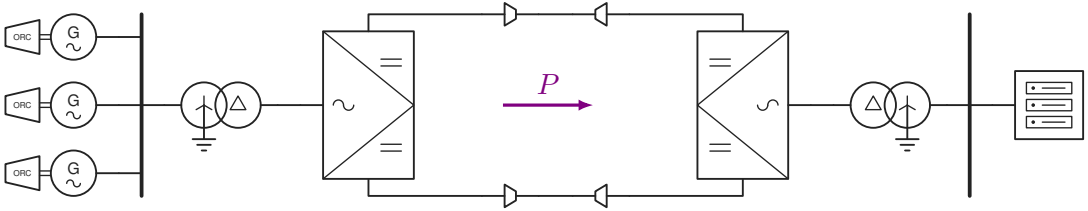


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System Overview

ORC generators feed a data centre through a point-to-point VSC HVDC link.



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SMP Level 1
Power path

From generation to load

A.c. collection, conversion to d.c., d.c. transmission, conversion back to a.c.

- Generation: three ORC sets on one busbar.
- Sending end: transformer YNd11, VSC converter.
- D.c. link: two poles, cable sealing ends at both ends.
- Receiving end: VSC converter, transformer, data centre.

Tag	Element
GEN1–3	ORC generator set
BUS1	Busbar, sending end
T1	Transformer YNd11
U1	VSC converter, sending end
CSE1–4	Cable sealing ends, d.c. link
U2	VSC converter, receiving end
T2	Transformer YNd11
BUS2	Busbar, receiving end
DATA	Data centre

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